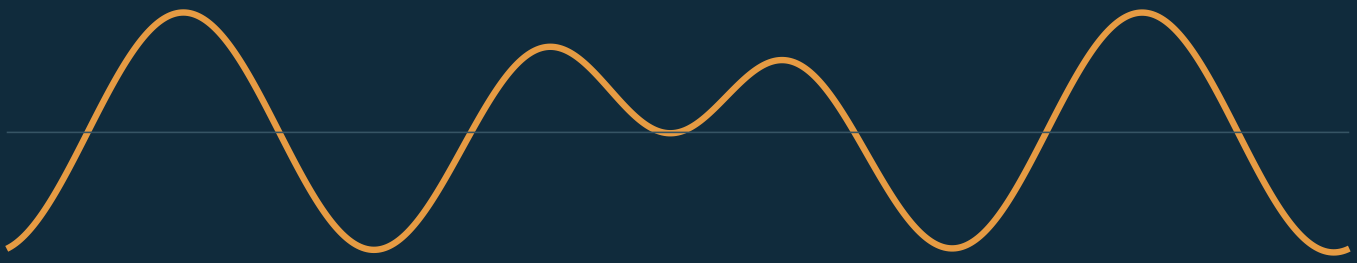
ORIGINAL MODEL / RESEARCH DEMONSTRATION

WiggleGPT

A language model with neurons
that learn to oscillate.



$$f(x) = \sin(\omega x + \varphi) \cdot \tanh(x) + b$$

Learnable frequency. Learnable phase. Learnable baseline.

123.7M
parameters

36,864
oscillating units

600k
pretraining updates

Prompts, unedited outputs and the training record. Fresh generations from the preserved original weights, accompanied by the model architecture and reproducible settings.

P.C. O'Brien

Independent Researcher | ORCID 0009-0007-3961-1182

26 October 2026 / Prepared with Codex

THE MODEL

Learning through oscillation

Original WiggleGPT v3. A GPT-style decoder with oscillating MLP activations.

WiggleGPT retains attention and the decoder structure while giving every MLP hidden feature its own learned frequency, phase and baseline. The response curve can change during training.

$$f(x) = \sin(\omega x + \varphi) \cdot \tanh(x) + b$$

ω: frequency φ: phase b: baseline

12
transformer layers

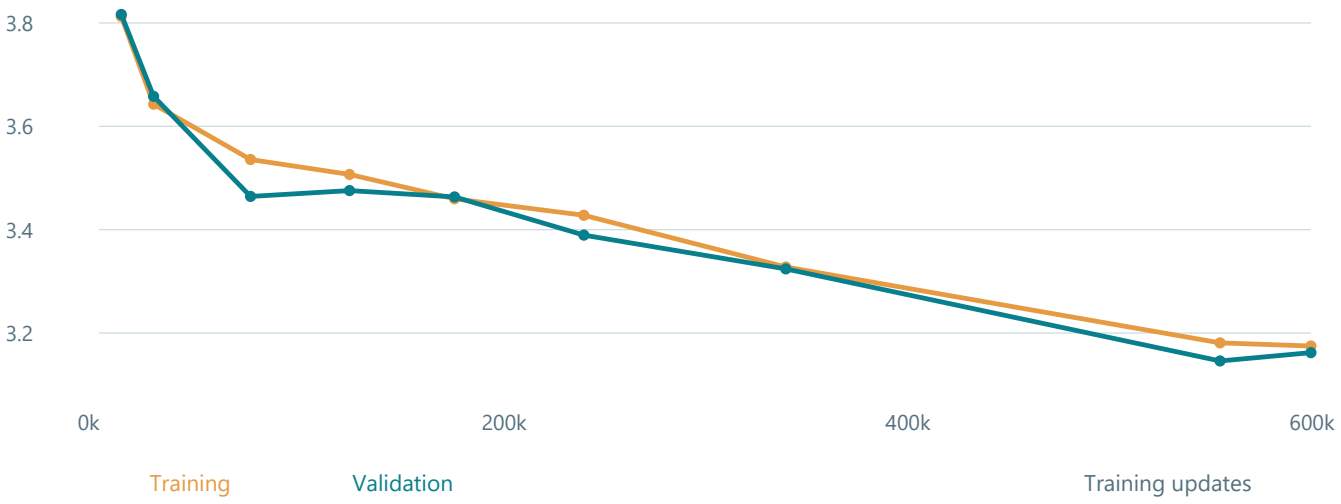
12
attention heads

768
embedding width

Each MLP expands 768 features to 3,072 oscillating units, then projects back to 768. The model uses RMSNorm, rotary position embeddings and tied input/output embeddings. Total parameters: 123,697,920.

PRETRAINING / OPENWEBTEXT

Validation loss 3.1621 at 600,000 updates



Selected recorded evaluations from 11k updates onward; points are joined for readability. The initial validation loss was 10.9413 at update 0. These are historical training records, not a new benchmark. [1]

Instruction tuning used SmolTalk2. The generations on the next two pages use the preserved native fine-tuned checkpoint saved at update 9,750. Its recorded loss field is 1.588843. [2]

SELECTED GENERATIONS

Narrative and scene writing

Exact prompts and complete captured replies from the original instruction-tuned weights.

A SHORT STORY

PROMPT

Write a short story about a lighthouse keeper who discovers a message in a bottle.

WIGGLEGPT OUTPUT

As the last remnants of sunlight faded, the lighthouse stood sentinel over the edge of the English countryside. The sounds of the sea and the creaking of old wooden beams stretched out before her like a whispered warning. The keeper, a young woman named Elara, had come to the aid of a shipwrecked sailor who had been stranded on the side of a dense forest for weeks. The message was etched in the glass: "Lift your spirit. We have found you."

A SCENE IN PROSE

PROMPT

Describe an old library on a rainy afternoon.

WIGGLEGPT OUTPUT

The old library that's been here for decades is a treasure trove of memories. As I gaze out the window, it's as if time has taken its toll on this small town like ours does. The air is thick with dust and cobwebs, and the only sounds are the creaking of wooden floorboards beneath my feet and the occasional distant hum of a nearby train.

Curated showcase. The complete prompt set and every captured output are included in the attached generation record.

SELECTED GENERATIONS

Summarisation and verse

Same checkpoint and sampling settings. No source notes or system prompt were supplied.

CONDENSING A PASSAGE

PROMPT

Summarise this in one sentence: A community garden opened on Saturday. Twenty volunteers planted herbs and vegetables. The garden will provide free food for local families.

WIGGLEGPT OUTPUT

Here's a rewritten version of the sentence: The community garden opened on Saturday, providing free food for local families.

A BRIEF VERSE

PROMPT

Write a poem about the sea.

WIGGLEGPT OUTPUT

The sea's secrets,
Its depths are a treasure trove o' gold.

Curated showcase. The complete prompt set and every captured output are included in the attached generation record.

RECORD AND SCOPE

How to read this demonstration

The examples show the original model generating language directly on a local GPU.

CAPTURE METHOD

Four replies were selected from 16 instruction prompts. Eight pretrained text continuations were also captured. Each prompt received one fixed-seed draw. The complete 24-prompt record includes every result.

Responses are reproduced without rewriting. The displayed replies use native user/assistant tags, with no history, system prompt, retrieval or external inference service. The sampler removes response-ending role markers.

SETTINGS FOR THE DISPLAYED OUTPUTS

Checkpoint	Original native fine-tune / update 9,750
Sampling	Temperature 0.7 / top-k 50 / top-p 1.0
Repetition penalty	1.15 / generated tokens / sign-aware scaling
Seed and allowance	Seed 42 for each prompt / up to 256 new tokens
Execution	NVIDIA GeForce RTX 5060 Ti / BF16 autocast

WHAT THE DEMONSTRATION ESTABLISHES

The model trained with oscillating neurons and generates narrative prose, summaries and verse in these examples. The full captures also contain inaccurate explanations, instruction failures, incorrect arithmetic and invalid code. This showcase is not an accuracy benchmark or a claim of dependable general-assistant performance.

SOURCE RECORD AND PUBLIC PROJECT

[1] Original log: Docs n' Logs/wiggleGPT trainlog4.md. [2] Original checkpoint metadata and model_bio.py, loaded directly. Checkpoint SHA-256 hashes, exact inputs, all outputs and settings are in WiggleGPT_Generation_Record.json, attached to this PDF and supplied alongside it.

Paper / DOI 10.5281/zenodo.17919011

Code / Eden-Eldith/WiggleGPT

Model card and weights / edeneldith/WiggleGPT

Architecture and research: P.C. O'Brien. Original framework: nanoGPT by Andrej Karpathy. Capture, selection and PDF preparation: Codex collaborating with the researcher. Public project links checked on 26 October 2026.

TASK-SPECIFIC FINE-TUNING

Learning the tasks

Separate adaptations of the original pretrained WiggleGPT. Reported accuracy uses held-out examples.

Each task starts from the original pretrained WiggleGPT. All model weights are adapted using classifier cross-entropy plus an auxiliary language-model loss (weight 0.5), with GPT-1-style task markers and linear classifiers.

SST-2

90.1%

after task training

+40.0 percentage points

RACE MIDDLE

59.5%

after task training

+32.7 percentage points

SAME-TASK CLASSIFIER COMPARISON

Task	Held-out rows	Before training	After training	Majority
SST-2	872	50.1% (437/872)	90.1% (786/872)	50.9%
RACE middle	1,436	26.8% (385/1,436)	59.5% (854/1,436)	27.0%

Before: original pretrained backbone with a newly initialized task classifier. After: the validation-selected task-trained checkpoint, using the same inputs and classifier format.

DATA USED FOR THE TASK ADAPTATIONS

SST-2: 67,349 training examples

872 validation examples used for checkpoint selection. Reported evaluation: 872 examples from validation.

RACE middle: 25,421 training examples

1,436 validation examples used for checkpoint selection. Reported evaluation: 1,436 examples from test.

Full SST-2 training and validation; full RACE-middle training, validation selection and held-out test. Historical GPT-1 figures use different training coverage and test sets.

Dataset sources: Stanford SST-2 / RACE (middle-school).

TRAINING AND CONTEXT

How the comparison was run

The recorded training recipe, checkpoint choice and historical reference.

SST-2

epochs: 3 / optimizer updates: 6,315 / microbatch: 16 / gradient accumulation: 2 / effective batch: 32

Stopped: completed 3 planned epochs. Selected update: 6,315. Recorded elapsed: 17m 13s.

Classifier: 768-to-2 linear.

RACE MIDDLE

epochs: 3 / optimizer updates: 2,385 / microbatch: 2 / gradient accumulation: 16 / effective batch: 32

Stopped: completed 3 planned epochs. Selected update: 2,385. Recorded elapsed: 1h 32m.

Classifier: shared 768-to-1 linear over four answers.

Learning rate: 6.25e-05 / token context: 512 / auxiliary LM weight: 0.5 / classifier dropout: 0.1

Full native backbone trained. Zero decay on oscillator and 1-D parameters. Linear GPT-1 warmup/decay, AdamW, gradient clip1.0. Original backbone dropout retained; classifier dropout0.1. Three unused padding-vocabulary slots initialized for task markers.

Checkpoint selection: highest validation accuracy, lower validation classification loss breaks ties.

Start: Original pretrained WiggleGPT / update 600,000. Original weights were preserved; each run contains copies and hashes of its trainer and model source.

THE GPT-1 REFERENCE

OpenAI reported 91.3% on SST-2 (GLUE test) and 62.9% on RACE-M. This WiggleGPT run reports SST-2 public validation. GPT-1 trained on combined RACE-middle/high data; this adaptation uses middle-school training data. The historical scores provide context with these differences in training coverage and evaluation split.

GPT-1 paper: Improving Language Understanding by Generative Pre-Training

EARLIER LIKELIHOOD PROBES / SEPARATE SCORING METHOD

The earlier fixed-slice answer-likelihood probes used no task classifier or task training. Their scoring method differs from the full held-out comparison above.

Task	Rows	Pretrained	General instruction-tuned
SST-2	100	53.0%	67.0%
RACE middle	40	27.5%	37.5%

SST-2: first 100 validation rows, sentiment-word likelihood. RACE: first 40 middle-school test rows, answer-token likelihood. General instruction tuning refers to the preserved SmolTalk2 checkpoint.

Full comparison and exact predictions: attached as WiggleGPT_Task_Finetuning_Comparison.json.

RECORDED DECISIONS

Examples from the held-out set

Saved model selections, exact task questions and clearly labelled context. Correct answers are dataset labels.

SST-2 / AFTER TRAINING: CORRECT

a gorgeous , high-spirited musical from india that exquisitely blends music , dance , song , and high drama .

A. negative
B. positive

Before: negative (incorrect)
After: positive (correct)
Correct answer: positive

Record: sst2/validation/15

RACE MIDDLE / AFTER TRAINING: CORRECT

Passage summary
Dan, a club doorman, grows tired of passers-by asking the time. He installs a large clock to stop the interruptions. Instead, people begin asking whether the clock is correct, so his attempted solution creates another recurring question.

Summary for display. The model received the original passage, question and choices.

What would be the best title for the passage?

A. Hardworking Dan
B. A Big Clock
C. Stupid Question
D. Boring People

Before: D (incorrect)
After: C (correct)
Correct answer: C

Record: race_middle/test/172

Examples were selected from the saved predictions. Passage summaries are display context; questions and choices are reproduced as tested. The preceding table reports every held-out row in each stated evaluation.